

Supervised Learning: Regressions, SVM & Decision Trees

Subject: CS601 | Unit 2: Supervised Methods

1. Executive Overview & Syllabus Scope

Mathematical formulation of Gradient Descent, Cost Functions, Regularization L1 Lasso, L2 Ridge, Maximum Margin Hyperplan

2. Key Architectural Invariants & Core Topics

â€¢ Lecture Modules & Key Topics

- Unit 2: Machine Learning Supervised Learning Foundations
- 1. Linear & Logistic Regression: Cost function formulation and Batch vs Stochastic Gradient Descent
- 2. Regularization & Generalization: L1 Lasso sparsity and L2 Ridge weight shrinkage techniques
- 3. Support Vector Machines SVM: Hyperplane geometry, Soft-margin C parameter, and Kernel trick RBF
- 4. Decision Trees: Shannon Entropy, Information Gain calculation, Gini Impurity, and Pruning strategies
- 5. Ensemble Techniques: Bagging Random Forest vs Boosting AdaBoost, XGBoost mechanics

3. University Exam Tips & High-Yield Checklist

- * Practice numerical problems and time complexity derivations under timed conditions.
- * Draw neat labeled block diagrams for full credit in subjective university evaluations.
- * Use AI_Book Quiz & Summary features to test active recall before mid-terms and finals.